Planar Steiner diversities need logarithmic ℓ_1 distortion

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Abstract

A diversity assigns a value to every finite set of points, extending a metric from pairs to sets. The Steiner diversity of a graph gives each set of vertices the length of the shortest connected subgraph that contains it. We show that the Steiner diversities of planar graphs cannot be embedded in ℓ_1 with bounded distortion, even though their shortest-path metrics can. The examples are the unit diamond graphs D_d, which are planar and have treewidth two. The graph D_d has $N_d = (2 \cdot 4^d + 4)/3$ vertices, and its Steiner diversity needs ℓ_1 distortion at least $(6d + 7 + 2/4^d)/9$. By an upper bound of Bryant and Tupper, the order $\log N_d$ is correct. Equivalently, D_d carries demands on vertex sets that pass every cut test but need congestion at least this large when routed by Steiner trees. The exact distortion is not determined for $d \geq 3$. Bounded literature searches found no earlier statement of the result, but its novelty is not established.

1 Introduction

A diversity assigns a value to every finite set of points, extending a metric from pairs to sets. The Steiner diversity of a graph gives each set of vertices the length of the shortest connected subgraph that contains it. We ask how well a single embedding in ℓ_1 can match these lengths for every vertex set at once, and show that on planar graphs no constant factor suffices.

Formally, a diversity on a finite set V is a function $\delta : 2^V \to [0, \infty)$ that vanishes exactly on sets of size at most one and satisfies

$$\delta(A \cup B) \leq \delta(A \cup C) + \delta(C \cup B) \qquad (C \neq \varnothing).$$

Bryant and Tupper introduced this framework [1] and developed its connection to hypergraph cuts and flows [2]. For a finite connected graph $G = (V, E)$ with positive edge lengths ℓ, its *graph Steiner diversity* is

$$\delta_{G,\ell}(A) = \min \left\{ \sum_{e \in E(H)} \ell(e) : H \subseteq G \text{ is connected and } A \subseteq V(H) \right\},$$

with value zero for $|A| \leq 1$. Any vertex of G may serve as a Steiner point, and every edge joins two vertices.

For $f : V \to \mathbb{R}^m$, the ℓ_1 diversity of f adds up how far each coordinate spreads over a set:

$$h_f(A) = \sum_{i=1}^{m} \left(\max_{v \in A} f_i(v) - \min_{v \in A} f_i(v) \right),$$

with $h_f(\varnothing) = 0$. Write $c_1(\delta)$ for the infimum of $D \geq 1$ over all dimensions m and maps f with $\delta(A) \leq h_f(A) \leq D\delta(A)$ for every $A \subseteq V$; rescaling f shows that this normalization loses nothing. For unit edge lengths we write $c_1(G)$.

For pairs of vertices, the question is settled on these graph classes. OpenAI's problem #089 papers prove that shortest-path metrics of planar graphs, and of graphs of bounded treewidth, embed in ℓ_1 with bounded distortion [5, 6]. For series-parallel graphs, which include the diamonds below, the sharp bound 2 is due to Chakrabarti, Jaffe, Lee and Vincent [7], as surveyed in [5, Section 1.2]. We show that once all vertex sets count, no constant bound holds, already for planar graphs of treewidth two.

Results. Theorem 1 gives an explicit lower bound for the unit diamond graphs at every depth. Its proof is a short certificate: one inequality, valid for every cut, on d nested vertex sets. Corollary 4 shows that the bound has the correct order, $\log N_d$, and Corollary 5 restates it as a gap between flows and cuts for demands on vertex sets. Propositions 6 and 7 show how the bound depends on edge lengths and that the certificate is the best of its shape. None of these results determines $c_1(D_d)$ for $d \geq 3$. The exact values in Section 4 are checks, not premises of the proof. Section 5 credits earlier work.

2 The diamond lower bound

Let D_0 be one edge. To obtain D_d from D_{d-1}, replace every edge by two internally disjoint paths of two edges each. Every final edge has unit length. Retain all earlier vertices, and let V_j be the vertex set present after generation j, viewed as a subset of V_d.

Theorem 1. *For every integer $d \geq 1$, the graph D_d has 4^d edges and $N_d = (2 \cdot 4^d + 4)/3$ vertices. It is planar and has treewidth exactly two. Its graph Steiner diversity satisfies*

$$c_1(D_d) \geq L_d := \frac{6d + 7 + 2/4^d}{9}.$$

Consequently, neither planar graphs nor graphs of treewidth two admit a uniform constant bound on ℓ_1 diversity distortion.

2.1 Cut representation and duality

For a cut $S \subseteq V$, let $\chi_S(A)$ be one if A meets both S and $V \setminus S$, and zero otherwise. Every coordinate-range ℓ_1 diversity on a finite set is a finite nonnegative sum

$$h(A) = \sum_S w_S \chi_S(A), \qquad w_S \geq 0.$$

Indeed, sort the distinct values of each coordinate and assign each consecutive gap to its upper threshold cut. A set crosses exactly the thresholds between its minimum and maximum. Conversely, the coordinate $w_S \mathbf{1}_S$ realizes $w_S \chi_S$.

Suppose nonnegative measures α, β on subsets of V satisfy

$$\sum_A \alpha_A \chi_S(A) \leq \sum_A \beta_A \chi_S(A) \quad \text{for every } S, \qquad \sum_A \beta_A \delta(A) = 1. \tag{1}$$

Then any $\delta \leq h \leq D\delta$ obeys

$$\sum_A \alpha_A \delta(A) \leq \sum_A \alpha_A h(A) \leq \sum_A \beta_A h(A) \leq D.$$

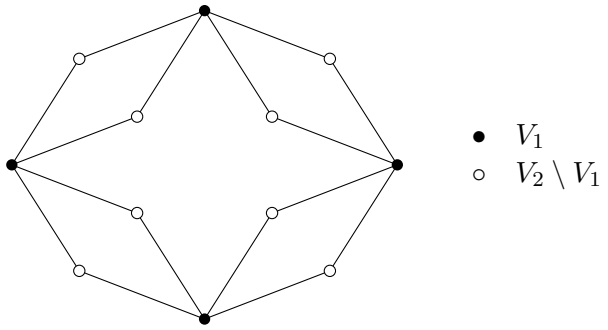


Figure 1: The unit diamond D_2. All twelve vertices form V_2; the four filled vertices form V_1. Their Steiner costs are respectively 11 and 6. The drawing does not encode edge lengths.

This is the dual-certificate principle used below. It applies in every allowed finite dimension.

2.2 Geometry and graph class

Each replacement multiplies the number of edges by four and adds two vertices per old edge. Thus

$$E_d = 4^d, \qquad N_d = 2 + 2\sum_{i=0}^{d-1} 4^i = \frac{2 \cdot 4^d + 4}{3}.$$

Each edge of D_j has a descendant gadget isomorphic to D_{d-j}. Distinct gadgets have disjoint edge sets and disjoint interiors, meeting only at old endpoints. Within a gadget the terminal distance is 2^{d-j}, and there are 2^{d-j} edge-disjoint terminal paths. Both assertions follow inductively: two smaller diamonds occur in series in each of two parallel branches. Distances double; corresponding paths concatenate in each branch, and the two branch collections are edge-disjoint.

Every edge of D_j, for $j \geq 1$, lies in the four-cycle replacing its parent edge. Hence D_j is connected and bridgeless, and each nontrivial cut crosses at least two edges.

Replacing each edge in a planar drawing inside a narrow neighborhood preserves planarity. For a width-two tree decomposition, start with the bag containing the original two endpoints. When xy is replaced through new vertices a, b, attach bags $\{x, y, a\}$ and $\{x, y, b\}$ to an existing bag containing x, y. Old edge coverage ensures such a bag exists. The new edges are covered, and the bags containing each vertex remain connected. Retain old bags after their edges disappear. Every bag has size at most three. Since D_d contains a cycle for $d \geq 1$, its treewidth is exactly two.

2.3 A simultaneous inequality for every cut

Set

$$a_j = 2^{d-j} \quad (1 \leq j < d), \qquad a_d = 2.$$

Lemma 2. *For every cut S of D_d,*

$$\sum_{j=1}^{d} a_j \chi_S(V_j) \leq |\partial_{D_d} S|. \tag{2}$$

Proof. Trivial cuts have both sides zero. For a nontrivial cut, let q be the first generation in $\{1, \ldots, d\}$ with $\chi_S(V_q) = 1$. It exists since V_d is the whole vertex set. By nesting, the indicators vanish before q and are one thereafter. The restriction to V_q is a nontrivial cut of D_q, so it crosses

at least two coarse edges. Their descendant gadgets are edge-disjoint. In each, S separates the endpoints and therefore crosses all 2^{d-q} edge-disjoint terminal paths. Consequently

$$|\partial_{D_d} S| \geq 2^{d-q+1} = \sum_{j=q}^{d} a_j = \sum_{j=1}^{d} a_j \chi_S(V_j).$$

This includes $q = d$ and $d = 1$. □

2.4 Exact Steiner costs of the generation sets

Lemma 3. *For $1 \leq j \leq d$,*

$$\delta_{D_d}(V_j) = 2^{d-j}(N_j - 1). \tag{3}$$

Proof. For the upper bound, choose a spanning tree of D_j and a shortest terminal path in each of its $N_j - 1$ edge gadgets. Their union is connected, contains V_j, and has $2^{d-j}(N_j - 1)$ edges.

For the lower bound, let H be any connected subgraph containing V_j. Retain a coarse edge of D_j whenever H connects its endpoints within its descendant gadget. The retained edges connect V_j: a simple H-path between old vertices, divided at successive old vertices, consists of passages through single gadgets between their endpoints. Different gadgets meet only at old endpoints. Each passage gives a retained coarse edge. At least $N_j - 1$ gadgets are therefore retained. Each costs at least 2^{d-j} edges, and their edge sets are disjoint. This proves the lower bound even when H uses arbitrary newer Steiner vertices. □

2.5 The certificate and its objective

Put $\alpha_{V_j} = a_j/4^d$ and put $\beta_{\{u,v\}} = 1/4^d$ on every final graph edge, with all other coefficients zero. Every edge pair has Steiner cost one, so $\sum_A \beta_A \delta(A) = 1$. Dividing (2) by 4^d gives every cut constraint in (1). By (3),

$$\begin{aligned}
\sum_A \alpha_A \delta(A) &= \frac{1}{4^d}\left[\sum_{j=1}^{d-1} \frac{4^{d-j}(2 \cdot 4^j + 1)}{3} + \frac{2(2 \cdot 4^d + 1)}{3}\right] \\
&= \frac{2(d-1)}{3} + \frac{1}{9}\left(1 - 4^{-(d-1)}\right) + \frac{4}{3} + \frac{2}{3 \cdot 4^d} \\
&= \frac{6d + 7 + 2/4^d}{9}.
\end{aligned}$$

The sum is empty when $d = 1$. The dual-certificate principle proves Theorem 1. Since N_d is proportional to 4^d, $L_d = (2/3)\log_4 N_d + O(1)$, which diverges.

3 Consequences and limits

Corollary 4. *The unit diamond diversities have $c_1(D_d) = \Theta(\log N_d)$. The lower bound is self-contained; the upper bound uses [3, Theorem 2].*

Proof. Bryant and Tupper prove an $O(\log n)$ upper bound for the Steiner diversity of an n-point metric space. Graph Steiner costs equal the Steiner costs in the shortest-path metric completion on the same vertex set. Replacing metric-tree edges by graph shortest paths gives one inequality; viewing a graph Steiner tree as a tree of metric edges gives the other. All vertices remain available as Steiner points. The cited theorem and Theorem 1 now give the claimed order. □

Corollary 5 (Set-demand flow–cut form)**.** *Give every edge of D_d capacity one and give terminal set V_j demand a_j, with all other demands zero. These demands satisfy every cut condition, but any fractional Steiner-tree routing of the full demands requires congestion at least L_d. In the notation of [2, Section 5],*

$$\text{MinHypCut} = 1, \qquad \text{MaxHSP} \le L_d^{-1}, \qquad \frac{\text{MinHypCut}}{\text{MaxHSP}} \ge L_d.$$

Proof. The cut condition is exactly (2). The cut ratio is boundary capacity divided by $\sum_j a_j \chi_S(V_j)$, for nonzero denominator. Every such ratio is at least one. A singleton vertex born in generation d has degree two and crosses only the demand set V_d, whose demand is two; its ratio is one.

For each j, distribute mass a_j over trees containing V_j. The sum of all edge loads is at least $\sum_j a_j \delta(V_j) = 4^d L_d$. Congestion ρ with unit capacities bounds that sum above by $\rho 4^d$, hence $\rho \ge L_d$. Bryant and Tupper's maximum concurrent packing variable f routes mass $f a_j$ under the original capacities. The same calculation gives $f 4^d L_d \le 4^d$. This verifies the stated normalization directly. Their Theorems 1–2 and 17–18 establish the general diversity/flow–cut correspondence; the calculation here supplies these particular demands. □

Proposition 6 (Secondary: length dependence)**.** *Common positive rescaling of all edge lengths preserves c_1. If the lengths on D_d satisfy $0 < m \le \ell(e) \le M$ and $R = M/m$, then*

$$c_1(D_d, \ell) \ge \max\{1, L_d/R\}.$$

There are also strictly positive length assignments on every underlying diamond for which $c_1 = 1$.

Proof. Common rescaling multiplies all Steiner costs and all embedding coordinates by the same factor. For bounded lengths, minimizing over the same connected subgraphs gives $m\delta_{\text{unit}} \le \delta_\ell \le M\delta_{\text{unit}}$. If $\delta_\ell \le h \le D\delta_\ell$, then h/m is noncontracting for δ_{unit} with distortion at most DR. Take infima and apply Theorem 1.

For the last assertion, choose a spanning tree T on N vertices, give its edges length one, and give every other edge length N. Every non-tree edge can be replaced by its T-path of at most $N-1$ edges. Thus every connected subgraph spanning A can be replaced by one in T of no greater cost, and $\delta_{G,\ell}(A) = \delta_T(A)$ for every A. The tree diversity is the sum of its edge-cut diversities, so it embeds exactly in ℓ_1. □

Proposition 7 (Secondary: optimum of the restricted method)**.** *On the unit diamond, fix $\beta_{\{u,v\}} = 1/4^d$ on every edge and require α to be supported on $V_1, \dots, V_d$. Among feasible nonnegative α in (1), the maximum objective is L_d, uniquely attained by the coefficients of Theorem 1.*

Proof. Write $x_j = 4^d \alpha_{V_j}$, $y_q = \sum_{j=q}^d x_j$, and $b_q = 2^{d-q+1}$. The all-cut feasible region is exactly

$$x_j \ge 0, \qquad y_q \le b_q \quad (1 \le q \le d).$$

Necessity follows from a singleton vertex born in generation q: its degree is b_q, and it crosses exactly the generation sets indexed at least q. For sufficiency, the proof of Lemma 2 bounds any cut first splitting V_q below by b_q boundary edges.

Let $c_j = \delta(V_j)$. Lemma 3 gives $c_1 > 0$ and $c_j - c_{j-1} = 2^{d-j}(4^j - 1)/3 > 0$ for $j \ge 2$. Summation by parts yields

$$4^{-d} \sum_{j=1}^d c_j x_j = 4^{-d} \left(c_1 y_1 + \sum_{j=2}^d (c_j - c_{j-1}) y_j \right).$$

Graph	Exact c_1	Graph	Exact c_1
Path, 8 vertices	1	D_2	17/8
Star, 8 vertices	1	Fan, 5 vertices	8/5
K_3	4/3	Fan, 8 vertices	7/4
K_4	3/2	Wheel, 5 vertices	5/3
K_5	8/5	Wheel, 8 vertices	38/21
K_6	5/3	Grid 2×3	5/3
K_8	7/4	Grid 2×4	7/4
C_4	3/2	Grid 2×5	9/5
C_5	8/5	Grid 2×6	11/6
C_6	5/3	Grid 3×3	11/6
C_7	12/7	Grid 3×4	29/15
C_8	7/4	Weighted 2-tree, seed 17	4/3
C_{10}	9/5	Weighted 2-tree, seed 31	14/11
C_{12}	11/6	Weighted 2-tree, seed 53	26/19

Table 1: Exact certificate values. Fans and wheels count all vertices, including the hub. The path has edge lengths $1, 2, 3$ repeated, the star has lengths $1, \dots, 7$, and each 2-tree has eight vertices and recorded integer lengths in $1, \dots, 9$. All other edges have unit length.

Every coefficient is positive, so the maximum requires $y_q = b_q$ for all q. These equalities are simultaneously feasible since b_q decreases. They give $x_j = b_j - b_{j+1} = 2^{d-j}$ for $j < d$ and $x_d = 2$, as claimed. □

This restricted optimality does not establish the full dual optimum or $c_1(D_d) = L_d$ for $d \geq 3$. Uniform lengths provide the witness family; Theorem 1 is not a lower bound for every positive-length diamond. The result concerns graph Steiner diversities with arbitrary graph vertices as Steiner points, and finite-dimensional coordinate-range ℓ_1 targets. It is not a claim about terminal-induced subgraphs, terminal-only Steiner trees, or the distortion of pairs alone.

4 Exact values for small graphs

Table 1 lists exact values of c_1 for 28 small graphs. Each value comes with a rational embedding and a rational dual certificate that attain it. The repository's checker builds D_1 and D_2, checks inequality (2) on every cut, computes their exact Steiner costs, and checks the objective formula for $d = 1, \dots, 12$. For each of the 28 graphs it computes every Steiner cost exactly, as the least spanning-tree cost over vertex sets containing the terminals, and checks every constraint of both certificates in rational arithmetic. These checks do not prove the theorem for larger depths.

The values were found by linear programming; the rational certificates make them exact, independent of solver tolerances. The equalities $c_1(D_1) = 3/2$ and $c_1(D_2) = 17/8$ match L_1 and L_2, but they do not show that $c_1(D_d) = L_d$ for larger d.

5 Earlier work

Bryant and Tupper introduced diversities [1], developed their cut representation and the correspondence between ℓ_1 distortion and flow–cut gaps [2], and proved the $O(\log n)$ upper bound for Steiner

diversities [3, Theorem 2]. We use the last result only for the upper half of Corollary 4 and have not re-proved it.

Jozefiak and Shepherd [4] note the general lower bound of order $\log n$ inherited from expander metrics; it says nothing about planar graphs. Gupta, Newman, Rabinovich and Sinclair [8] show that some series-parallel metrics need distortion of order $\log n$ in distributions over dominating trees. That is a statement about metrics and a different target, but it suggests a reading of Theorem 1: for Steiner diversities of diamonds, ℓ_1 does no better than random trees, up to a constant factor.

The author's earlier paper [9] studies Steiner diversities of hypergraphs, whose edges may contain more than two vertices, through an affine construction. The present examples are ordinary graphs.

Bounded searches of diversity and flow–cut terminology, and of papers citing [2] and [4], found no earlier statement of Theorem 1 or of its certificate. This does not establish priority. The repository records each search and its limits.

AI assistance

AI systems helped find the result, attack and check the proof, verify the finite certificates, extract bibliographic data and draft this paper, under the author's direction. A separate AI review checked the full proof. Such reviews are not peer review or a formal proof. The author is responsible for the claims and their presentation.

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